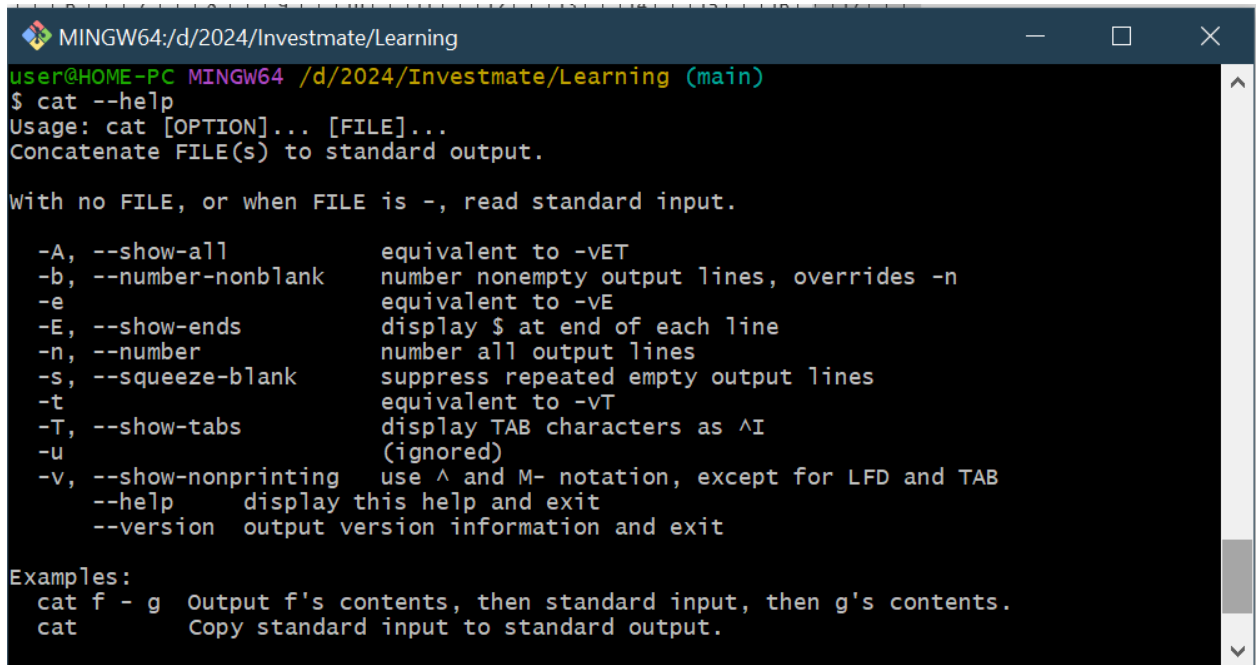


Linux terminal (command cat)

Команда **cat** одна из самых полезных утилит, которые обязательно стоит выучить. Являясь сокращением английского слова «concatenate» (конкатенация), она позволяет создавать, объединять, а также выводить содержимое файлов в командной строке или в другом файле. Сегодня мы рассмотрим общие способы использования команды Linux cat, объясняя ее особенности.

- Чтобы найти все доступные опции данной команды, просто введите в терминале **cat --help** ознакомьтесь с ними.



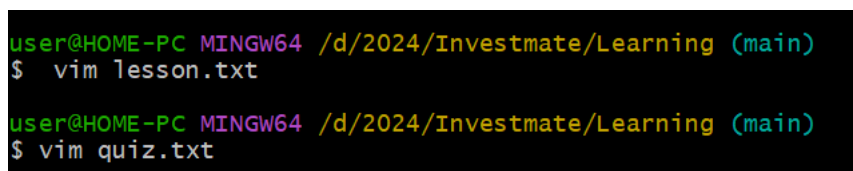
```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat --help
Usage: cat [OPTION]... [FILE]...
Concatenate FILE(s) to standard output.

With no FILE, or when FILE is -, read standard input.

-A, --show-all           equivalent to -vET
-b, --number-nonblank     number nonempty output lines, overrides -n
-e, --equivalent to -vE
-E, --show-ends          display $ at end of each line
-n, --number             number all output lines
-s, --squeeze-blank      suppress repeated empty output lines
-t, --equivalent to -vT
-T, --show-tabs          display TAB characters as ^I
-u, --ignored
-v, --show-nonprinting   use ^ and M- notation, except for LFD and TAB
--help                  display this help and exit
--version               output version information and exit

Examples:
cat f - g  Output f's contents, then standard input, then g's contents.
cat        Copy standard input to standard output.
```

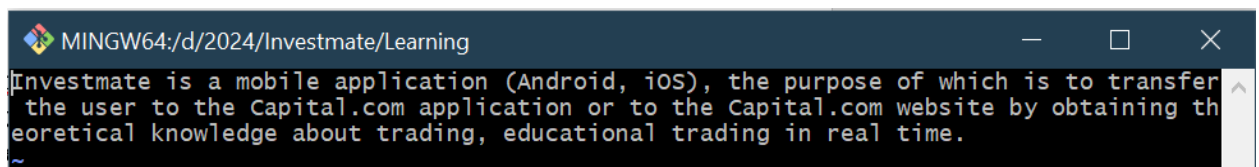
Далее для применения данной утилиты на практике, откройте локальный репозиторий Investmate, запустите Git Bash. Перейдите в папку Learning. С помощью команды vim добавьте текст в следующие файлы: lesson.txt и quiz.txt.



```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ vim lesson.txt

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ vim quiz.txt
```

В текстовый файл **lesson.txt** введите **Investmate is a mobile application (Android, iOS), the purpose of which is to transfer the user to the Capital.com application or to the Capital.com website by obtaining theoretical knowledge about trading, educational trading in real time.**



```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat lesson.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer
the user to the Capital.com application or to the Capital.com website by obtaining th
eoretical knowledge about trading, educational trading in real time.
```

В текстовый файл **quiz.txt** введите **The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.**

```
MINGW64:/d/2024/Investmate/Learning
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.
```

- В прошлой презентации мы с вами упоминали команду **cat**, одной из функций которой является *создание файла и введение в него текста*. (Для этого мы использовали оператор `>` redirect, чтобы перенаправить текст в файл (пример, `cat > lesson.txt`). После того как файл создан, его можно наполнять текстом. Если вы хотите добавить несколько строк, просто нажимайте Enter в конце каждой строки. Когда завершите, нажмите CTRL+D, чтобы выйти из файла. Создание файла проверьте командой терминала **ls**.)
- Рассмотрим также функцию *перенаправления вывода в другой файл*, используя параметр `>`.

c

Если lesson2.txt не существует, то команда создаст его или перезапишет контент из lesson.txt в существующий с таким же именем lesson2.txt.

- Рассмотрим следующую функцию команды **cat**, такую как *объединение файлов*:

cat lesson.txt quiz.txt > combined.txt

s

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat lesson.txt quiz.txt > combined.txt

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ ls -la
total 7
drwxr-xr-x 1 user 197121  0 Apr  9 01:45 ./
drwxr-xr-x 1 user 197121  0 Mar 19 02:24 ../
-rw-r--r-- 1 user 197121 487 Apr  9 01:45 combined.txt
-rw-r--r-- 1 user 197121 242 Apr  9 01:34 lesson.txt
-rw-r--r-- 1 user 197121 245 Apr  9 01:38 quiz.txt
-rw-r--r-- 1 user 197121  0 Mar 18 15:45 tradingtip.json
-rw-r--r-- 1 user 197121  0 Mar 18 15:45 trendpredictor.json
```

t

Мы видим, что данная команда соединяет содержимое lesson.txt и quiz.txt и сохраняет результат в combined.txt.

- С помощью еще одной функции команды **cat** *просмотр содержимого файла* мы убедимся, что команда выводит содержимое файла в терминал.

cat combined.txt

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat combined.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer the user to the Capital.com application or to the Capital.com website by obtaining theoretical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.
```

- Используем еще одну функцию команды **cat** *добавление текста в файл с опцией >>*, данная команда добавляет введенный текст в конец файла (без перезаписи).

cat >> combined.txt

Скопируйте следующий текст **With access to unlimited knowledge, you can start with the basics and work your way into the intricacies of the investment world. You will learn what CFDs are, how to trade CFDs on stocks, commodities and indices, learn to understand charts and build your own trading strategy.** и введите его в терминале.

Завершите, используя Ctrl+D

```

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat combined.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer
the user to the Capital.com application or to the Capital.com website by obtaining th
eoretical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD tr
ading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a gl
ossary of terms and tailored financial content - all in one application.

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat >> combined.txt
With access to unlimited knowledge, you can start with the basics and work your way into
the intricacies of the investment world. You will learn what CFDs are, how to trade CFD
s on stocks, commodities and indices, learn to understand charts and build your own trad
ing strategy.

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)

```

Выведем еще раз в терминал содержимое файла combined.txt

cat combined.txt

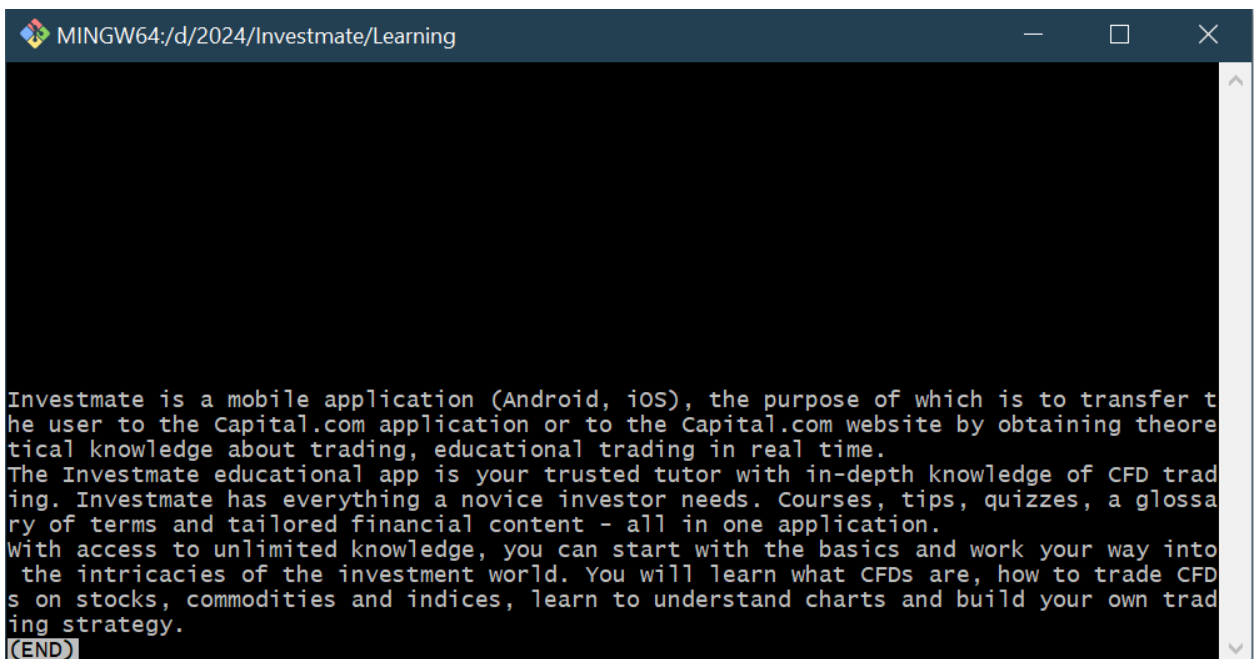
```

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat combined.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer t
he user to the Capital.com application or to the Capital.com website by obtaining theore
tical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trad
ing. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossa
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With access to unlimited knowledge, you can start with the basics and work your way into
the intricacies of the investment world. You will learn what CFDs are, how to trade CFD
s on stocks, commodities and indices, learn to understand charts and build your own trad
ing strategy.

```

- Для *удобного просмотра длинных файлов*, используйте параметр | **less**, который позволит скроллить выходные данные по одному скрину за раз.

cat combined.txt | less



```

MINGW64:/d/2024/Investmate/Learning
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer t
he user to the Capital.com application or to the Capital.com website by obtaining theore
tical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trad
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With access to unlimited knowledge, you can start with the basics and work your way into
the intricacies of the investment world. You will learn what CFDs are, how to trade CFD
s on stocks, commodities and indices, learn to understand charts and build your own trad
ing strategy.
(END)

```

- Для *отображения содержимого более одного файла*, введите в терминале следующую команду

cat *.txt выведет содержимое всех текстовых файлов в терминале

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat *.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer the user to the Capital.com application or to the Capital.com website by obtaining theoretical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.
With access to unlimited knowledge, you can start with the basics and work your way into the intricacies of the investment world. You will learn what CFDs are, how to trade CFDs on stocks, commodities and indices, learn to understand charts and build your own trading strategy.
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer the user to the Capital.com application or to the Capital.com website by obtaining theoretical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
```

- Есть еще одна полезная функция команды **cat** – *нумерация строк при выводе* (удобно для анализа логов). Реализует эту функцию опция **-n**. Проверим отображение содержимого файла с пронумерованными строками.

cat -n combined.txt

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat -n combined.txt
 1 Investmate is a mobile application (Android, iOS), the purpose of which is to transfer the user to the Capital.com application or to the Capital.com website by obtaining theoretical knowledge about trading, educational trading in real time.
 2 The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.
 3 With access to unlimited knowledge, you can start with the basics and work your way into the intricacies of the investment world. You will learn what CFDs are, how to trade CFDs on stocks, commodities and indices, learn to understand charts and build your own trading strategy.
```

- Чтобы вывести *номер каждой строки, не учитывая пустых*, используйте опцию **-b**. **cat -b** - нумерует только не пустые строки. Но помните, что параметр **-b** отменяет опцию **-n**
- С помощью команды **cat** можно обозначить конец строки. Маркер конца строки – символ \$. Чтобы реализовать эту функцию, используйте параметр **-E**

cat -E combined.txt

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat -E combined.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer the user to the Capital.com application or to the Capital.com website by obtaining theoretical knowledge about trading, educational trading in real time. $
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trading. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossary of terms and tailored financial content - all in one application.$
With access to unlimited knowledge, you can start with the basics and work your way into the intricacies of the investment world. You will learn what CFDs are, how to trade CFDs on stocks, commodities and indices, learn to understand charts and build your own trading strategy. $
```

- Команда **cat -s** удаляет повторяющиеся пустые строки.
- Команда **cat -v** отображает непечатаемые символы.
- Командой **cat** также можно вывести несколько файлов подряд:

cat lesson.txt quiz.txt combined.txt

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ cat lesson.txt quiz.txt combined.txt
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer t
he user to the Capital.com application or to the Capital.com website by obtaining theore
tical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trad
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Investmate is a mobile application (Android, iOS), the purpose of which is to transfer t
he user to the Capital.com application or to the Capital.com website by obtaining theore
tical knowledge about trading, educational trading in real time.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trad
ing. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossa
ry of terms and tailored financial content - all in one application.
With access to unlimited knowledge, you can start with the basics and work your way into
the intricacies of the investment world. You will learn what CFDs are, how to trade CFD
s on stocks, commodities and indices, learn to understand charts and build your own trad
ing strategy.

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$
```

- Чтобы посмотреть *содержимое файла в обратном порядке*, начиная с последней строки и заканчивая первой, просто используйте команду tac (команду cat, написанную в обратном порядке)

tac combined.txt

```
user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ tac combined.txt
With access to unlimited knowledge, you can start with the basics and work your way into
the intricacies of the investment world. You will learn what CFDs are, how to trade CFD
s on stocks, commodities and indices, learn to understand charts and build your own trad
ing strategy.
The Investmate educational app is your trusted tutor with in-depth knowledge of CFD trad
ing. Investmate has everything a novice investor needs. Courses, tips, quizzes, a glossa
ry of terms and tailored financial content - all in one application.
Investmate is a mobile application (Android, iOS), the purpose of which is to transfer t
he user to the Capital.com application or to the Capital.com website by obtaining theore
tical knowledge about trading, educational trading in real time.

user@HOME-PC MINGW64 /d/2024/Investmate/Learning (main)
$ |
```